

JIACHENG GAO

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RESEARCH INTEREST

Zero-knowledge Proofs, Secure Multiparty Computation, Cryptography, Security

EDUCATION

Nanjing University, Nanjing, China 09/2024 - 06/2027

Major: Master of Science in Computer Science and Technology

Nanjing University, Nanjing, China 09/2020 - 06/2024

Major: Bachelor of Science in Computer Science and Technology | **GPA: 4.50/5.0**

Related Course: Discrete Mathematics (98), Elements of Cryptography (96), Graph Theory and Algorithms (98), Formal Languages and Automata (96), Combinatorics (94), Computational Complexity (97), Advanced Algorithm (91)

RESEARCH EXPERIENCE

Intern, Research Group of Jiaheng Zhang, NUS. Jul 2025 - Feb 2026

- Designed and implemented a zero-knowledge proof framework for verifiable CNN training based on coding-based multilinear PCS (Orion) and the sum-check protocol.
- Proposed a new batch-commitment scheme for multilinear polynomials, reducing proof size and amortizing commitment costs.
- Developed a ZKP protocol for training and inference of gradient-boosting decision trees (GBDT) using univariate PCS (KZG) and lookup arguments.
- Proposed new batching tools for univariate polynomial commitments, and new approaches to commit GBDT with constant training/inference proof size.
- Co-authored two papers; one under review, the other accepted to ACM CCS 26.

M.S. Researcher, Research Group of Yuan Zhang, NJU. Sep 2024 - present

- Proposed a generic transformation from permutation protocols to shuffle protocols with linear online communication and computation complexity, whereas prior work incurs quadratic complexity.
- Optimized the shuffle protocol under Shamir secret sharing and implemented a prototype system with empirical evaluation.
- Designed MPC protocols for small-element operations, including one-hot encoding, comparison, digit decomposition, and sorting.
- Co-authored three papers, accepted to Asiacrypt 26, TIFS, JCS, respectively.

PUBLICATIONS

Zero-Knowledge Proofs for Gradient Boosted Decision Trees

Jiacheng Gao, Wenjie Qu, Yuan Zhang, Sheng Zhong, Jiaheng Zhang

ACM CCS 26; [ePrint Link](#)

Efficient SNARK for Convolutional Neural Network Training

Wenjie Qu*, Jiacheng Gao*, Jiaheng Zhang

Manuscript in preparation

UFOs: An Ultra-fast Toolkit for Multiparty Computation of Small Elements

Jiacheng Gao, Moyang Xie, Yuan Zhang, Sheng Zhong

Transactions on Information Forensics and Security (TIFS); [ePrint Link](#)

SLIDE: Shuffle Shamir Secret Shares Uniformly with Linear Online Communication and Guaranteed Output Delivery

Jiacheng Gao, Moyang Xie, Yuan Zhang, Sheng Zhong

Journal of Computer Security (JCS); [ePrint Link](#)

Free Linear Online Phase for Secure Multiparty Shuffle

Jiacheng Gao, Yuan Zhang, Sheng Zhong, Changyu Dong

ASIACRYPT 2026; [ePrint Link](#)

Revisiting Privacy-Preserving Min and k-th Min Protocols for Mobile Sensing

Jiacheng Gao, Yuan Zhang, Sheng Zhong

IEEE Transactions on Dependable and Secure Computing (TDSC), 2023. [Link](#)

HONORS & AWARDS

Outstanding Graduate of Nanjing University	2024
Elite Program Scholarship, Grand Award	2023
Elite Program Scholarship, Excellence Award	2021 and 2022
People Scholarship, Third Prize	2021 and 2022
National Olympiad in Informatics (Zhejiang Province), First Prize (Top 1%)	2019

SKILLS

Programming: C++, Python

Math: Abstract Algebra, Information Theory, Complexity Theory